

Normalization and scaling

COURSE ON SCRNA-SEQ DATA ANALYSIS

Reducing technical differences for meaningful comparison of expression profiles

Lucie Pfeiferova 7.-10. September 2026

Institute of Molecular Genetics of the Czech Academy of Sciences

Motivation and Learning objectives

- Normalization reduces technical differences so that differences between cells are not technical but biological, allowing meaningful comparison of expression profiles between cells.
- Distinguish Normalization, Transformation, and Scaling
- Identify and apply Normalization techniques

Source of differences

- **Biological**

- Cell subtype differences - size and transcriptional activity, variation in gene expression

- **Technical: scRNA data is inherently noisy**

- Low mRNA content per cell
 - Cell-to-cell differences in mRNA capture efficiency
 - Variable sequencing depth
 - PCR amplification efficiency

Raw counts problematic

1. **Library size bias** → deeper sequenced cells dominate
2. **Zero-inflation** → most genes = 0
3. **High dynamic range**

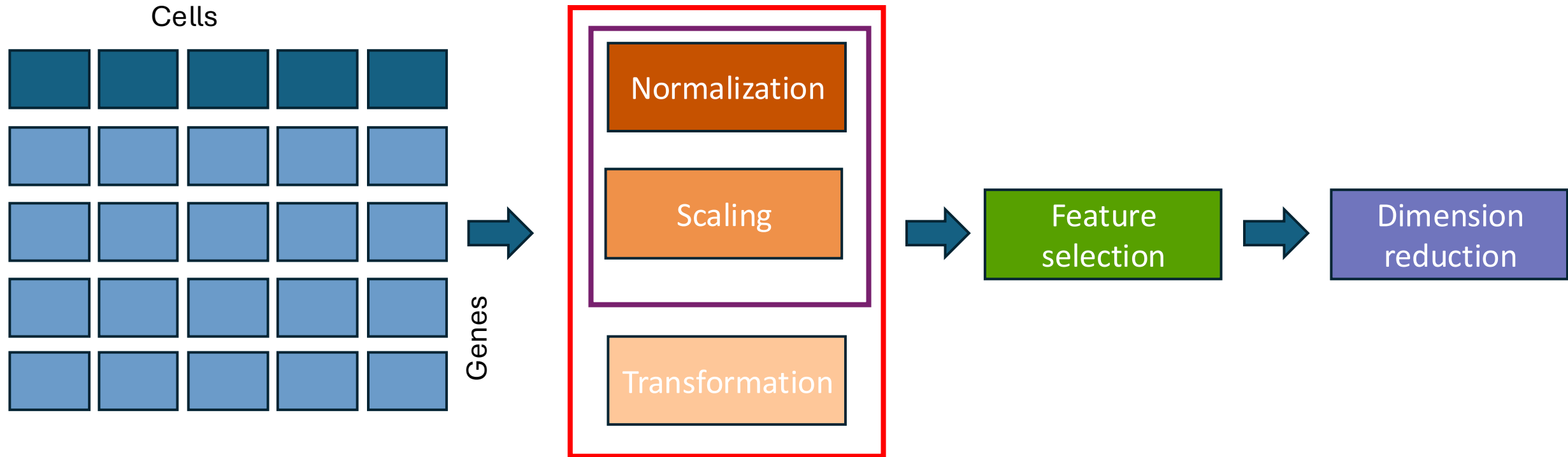
Normalization, transformation, scaling

Normalization = correct for sequencing depth
→ "Make cells comparable."

Scaling = equalize gene variance
→ "Make genes comparable."

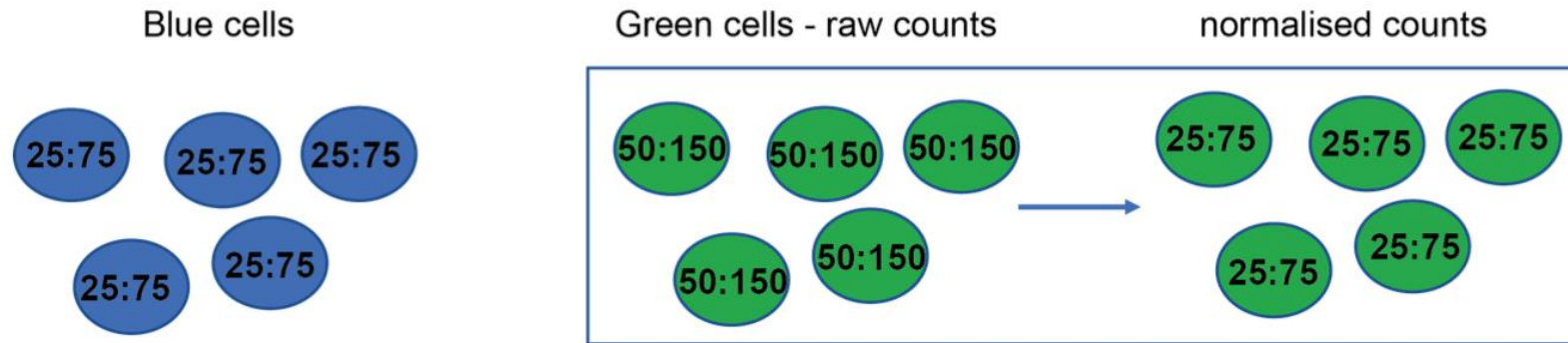
Transformation = stabilize distribution
→ "Make the numbers behave statistically."

Normalization process in scRNA-seq



Normalization process in scRNA-seq

1. Remove sequencing-depth (library-size) differences so that cells are comparable.
2. Scale to common factor



Normalization process in scRNA-seq

1. Remove sequencing-depth (library-size) differences so that cells are comparable.
2. Scale to common factor
3. Transform values (with log)

	Raw data				Log ₂ transform		
	Cell Type A	Cell Type B	Δ		Cell Type A	Cell Type B	Δ
Gene 1	1	2	1		0	1	1
Gene 2	100	200	100		6.64	7.64	1

SCTransform

1. Model fitting:

- Fit a negative binomial generalized linear model (NB-GLM) for each gene.
- Total UMI count per cell is used as a covariate (proxy for sequencing depth).

2. Parameter regularization:

- Intercept, slope, and dispersion parameters are regularized based on their relationship with gene mean.
- Prevents overfitting and stabilizes parameter estimates, especially for lowly expressed genes.

3. Variance-stabilizing transformation:

- Use the regularized model to compute Pearson residuals.
- These residuals represent normalized, variance-stabilized expression values.

SCTransform

1. Model fitting:

- Fit a negative binomial model
- Total UMI count per cell

2. Parameter regularization:

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- Relationship with gene expression
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Raw data

Gene 1

Gene 2

Cell Type A	Cell Type B	Δ
1	2	1
100	200	100

$\log_2$ transform

Cell Type A	Cell Type B	Δ
0	1	1
6.64	7.64	1

Pearson residuals

Gene 1

Gene 2

Cell Type A (50%)	Cell Type B (50%)	Δ
0.816	1.63	0.814
8.16	16.3	8.14

Summary

Step	What it does	Why it's needed	Typical methods
Normalization	Adjusts library size / sequencing depth differences between cells	Cells have different total RNA counts; normalization puts them on the same scale	CPM/TPM, Size-factor methods (e.g., scran), SCTransform's model-based approach
Scaling	Adjusts per-gene variance across genes or cells	Prevents highly expressed genes from dominating analyses such as PCA	Z-score scaling, regression of covariates
Transformation	Changes the distribution of gene expression values	RNA counts are skewed; transformations stabilize variance and make data more normally distributed	log1 p, variance stabilizing transformation (VST), SCTransform

Bonus – normalization by deconvolution

- Why deconvolution?
- Single cells have low UMI counts, making size factor estimation unstable.
- Many genes show zero expression, causing unreliable per-cell normalization.
- Deconvolution improves accuracy by borrowing information across cells

Bonus – normalization by deconvolution

1. Pooling:

- Group cells into pseudo-bulk pools to create higher-coverage “pseudo-samples.”
- This reduces sparsity and increases signal for size factor estimation.

2. Compute pool-based size factors:

- Estimate size factors on pooled data, where zeros are less problematic.

3. Deconvolution:

- Mathematically solve for individual cell size factors from overlapping pools.
- Ensures cell-specific normalization factors consistent across all pools.

4. Normalize counts:

- Divide each cell's gene counts by its deconvolved size factor.

Bonus – normalization by deconvolution

1. Pooling:

- Group cells samples."
- This reduces

2. Compute po

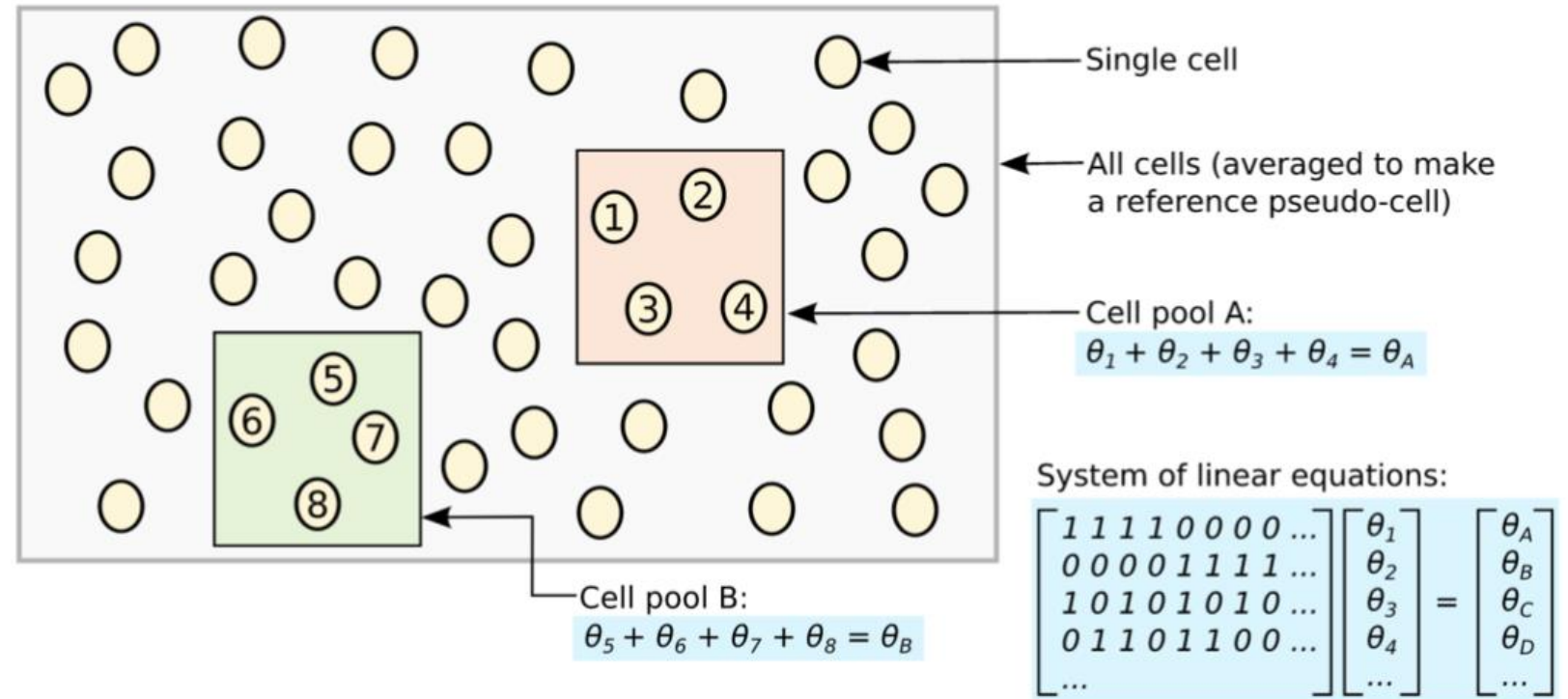
- Estimate size

3. Deconvoluti

- Mathematical
- Ensures cell-s

4. Normalize c

- Divide each cell's gene counts by its deconvolved size factor.



- <https://sib-swiss.github.io/single-cell-r-training/>
- Amezquita, R. A., Lun, A. T. L., Hicks, S. C., Marini, F., et al. (2020). Orchestrating single-cell analysis with Bioconductor. Nature Methods, 17(2), 137–145.
- <https://bioconductor.org/books/3.14/OSCA.basic/>
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- Hafemeister C, Satija R (2019). “Normalization and variance stabilization of single-cell RNA-seq data using regularized negative binomial regression.” Genome Biology, 20, 296. doi:10.1186/s13059-019-1874-1, <https://doi.org/10.1186/s13059-019-1874-1>.